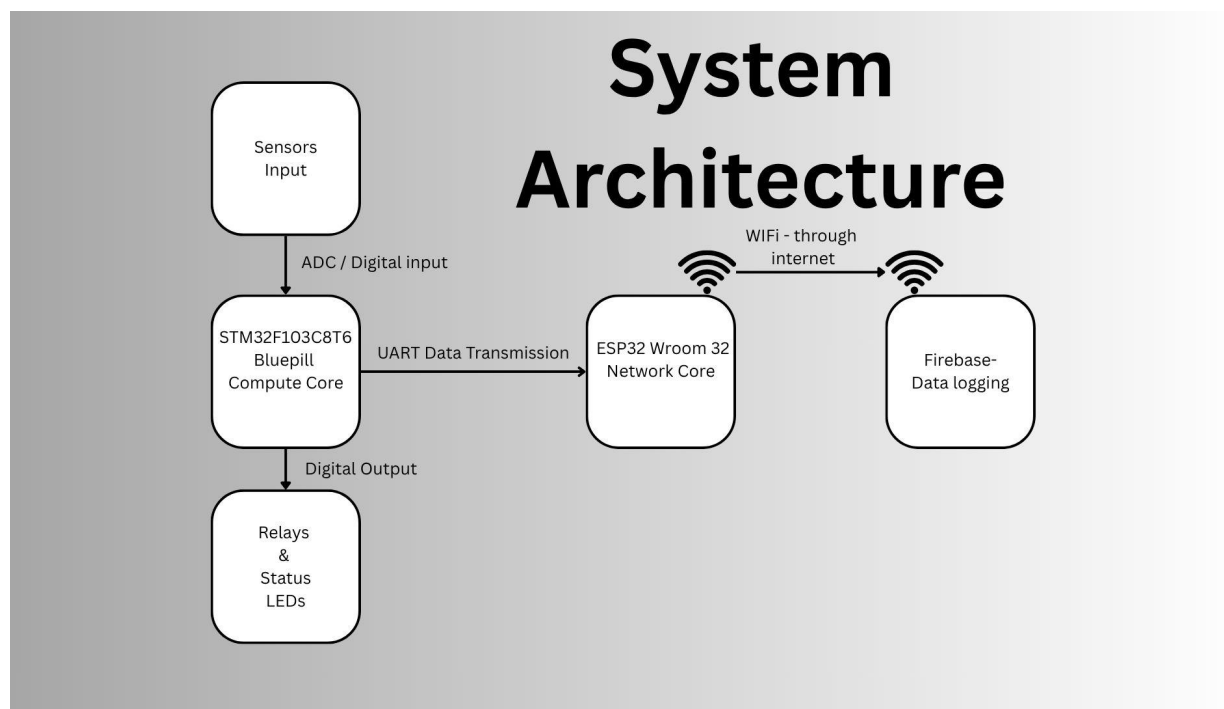


Task 9 – Solar Pump Monitoring Controller

STM32 planning Doc Prototype V1.0

1. Objective

The objective of this project is to design firmware for a Solar Pump Monitoring Controller using STM32F103 (Blue Pill). The controller monitors solar voltage, Battery voltage, tank level, pump feedback, Pump current and dry-run conditions. Based on the monitored inputs, the controller controls the pump, charging indication, alarm indication, buzzer notification and Status indication LEDs.



2. System Monitoring Parameters

The controller continuously monitors the following parameters:

Monitoring Parameter	Signal Type
Solar Panel Voltage	Analog
Battery Voltage	Analog
Pump Current	Analog
Tank Level	Analog
Pump Feedback	Digital
Dry Run Input	Digital

3. System Control Outputs

Output	Function
Pump Relay	Controls Pump Operation
Charging Relay	Indicates Battery Charging
Alarm Relay	Indicates Fault Condition
Buzzer	Audible Fault Indication
Status LED	System Health Indication

For prototype implementation, LEDs are used instead of relays.

4. Hardware Platform

Controller

- STM32F103C8T6 Blue Pill

Sensors

- Potentiometer (Solar Voltage Simulation)
- Potentiometer (Battery Voltage Simulation)
- Potentiometer (Pump Current Simulation)
- Water Level Sensor
- Hall Sensor (Pump Feedback)
- IR Sensor (Dry Run Detection)

Outputs

- Pump LED
- Charging LED
- Alarm LED
- Buzzer
- Onboard Status LED

5. GPIO Allocation Table

STM32 Pin	Function
PA0	Solar Voltage ADC
PA1	Battery Voltage ADC
PA2	Pump Current ADC
PA3	Tank Level ADC - Water Level Sensor
PA4	Pump Feedback Input - Hall Sensor
PA5	Dry Run Input - IR Sensor
PB0	Pump LED
PB1	Charging LED
PB10	Alarm LED

STM32 Pin	Function
PB11	Buzzer
PC13	Status LED
PA9	UART TX
PA10	UART RX - not used in this prototype V1.0

7. Battery Protection Logic

Battery voltage is continuously monitored.

Condition:

Battery Voltage < 2.3V

Actions:

- Pump OFF
- Alarm ON
- Buzzer ON
- Charging Indicator ON

Purpose:

Protect battery from deep discharge.

```
// Charging Logic
if(solar_voltage > SOLAR_CHARGE_VOLTAGE && battery_voltage < 3.2)
{
    LED_ON(GPIOB, GPIO_PIN_1);          // Charging LED
    LED_ON(GPIOC,GPIO_PIN_13);
    chargingStatus = 1;
}
else
{
    LED_OFF(GPIOB, GPIO_PIN_1);
    LED_ON(GPIOC,GPIO_PIN_13);
    chargingStatus = 0;
}
```

```
else
{
    // Low Battery Fault

    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_ON(GPIOB, GPIO_PIN_10);

    BUZZER_ON();
    LED_OFF(GPIOC,GPIO_PIN_13);
}
```

8. Pump Control Logic

Pump Start Conditions

The pump is allowed to start only if:

- Tank Level < 20%
- Battery Voltage > 2.3V
- Dry Run Input = OFF
- Pump Feedback Available

Actions:

- Pump ON
- Status LED ON

Pump Stop Conditions

The pump is stopped when:

- Tank Level > 80%
- Dry Run Detected
- Battery Voltage Low
- Pump Feedback Lost

Actions:

- Pump OFF
- Alarm ON
- Buzzer ON

```
if(tank_level < TANK_LOW_LEVEL)
{
    if(battery_voltage > BATTERY_MIN_VOLTAGE)
    {
        if(dryRun == 1)
        {
            // Start Pump
            LED_ON(GPIOB, GPIO_PIN_0);
            LED_ON(GPIOC,GPIO_PIN_13);
            pumpStatus = 1;
            //HAL_Delay(2000);
            if(pumpFeedback)
            {
                // Normal Operation
                LED_OFF(GPIOB, GPIO_PIN_10);
                BUZZER_OFF();
                pumpStatus = 1;
                LED_ON(GPIOC, GPIO_PIN_13);
            }
            else
            {
                // Pump Feedback Fault
                LED_OFF(GPIOB, GPIO_PIN_0);
                pumpStatus = 0;
                LED_ON(GPIOB, GPIO_PIN_10);
                BUZZER_ON();
                LED_OFF(GPIOC,GPIO_PIN_13);
            }
        }
        else
        {
            // Dry Run Fault
            LED_OFF(GPIOB, GPIO_PIN_0);
            pumpStatus = 0;
            LED_ON(GPIOB, GPIO_PIN_10);
        }
    }
}
```

```

        BUZZER_ON();
        LED_OFF(GPIOC,GPIO_PIN_13);
    }
}
else
{
    // Low Battery Fault
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_ON(GPIOB, GPIO_PIN_10);
    BUZZER_ON();
    LED_OFF(GPIOC,GPIO_PIN_13);
}
}
else if(tank_level >= TANK_HIGH_LEVEL)
{
    // Tank Full
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_OFF(GPIOB, GPIO_PIN_10);
    BUZZER_OFF();
    LED_ON(GPIOC,GPIO_PIN_13);
}
}

```

8. UART Communication

UART Interface:

- USART1
- Baud Rate: 115200 bps

Data transmitted every 3 seconds.

Example Data Packet:

SV=2.45,BV=2.80,PC=1.20,TL=45.0,PF=1,DR=0,PS=1,CS=0

Where:

- SV = Solar Voltage
- BV = Battery Voltage
- PC = Pump Current
- TL = Tank Level

- PF = Pump Feedback
- DR = Dry Run Status
- PS = Pump Status
- CS = Charging Status

9. Development Tools

- STM32CubeMX
- STM32CubeIDE
- Embedded C
- ST-Link V2 Programmer

10. Working Prototype Videos link:

<https://drive.google.com/drive/folders/1xAr9otawPFD658yjljqZsjYmVOqOD3Ba?usp=sharing>

Source Codes: Github link:

<https://github.com/arivenkkataram-asj/Solar-Pump-Monitoring-Controller.git>

11. Working Prototype Images:

